

Sylow Plan

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November 2025

Definitions

- G be a finite group, where $|G| = p^n m$ and $\gcd(p, m) = 1$
- p is a prime
- $\text{Syl}_p(G)$ is the set of Sylow p -subgroups
- $X = \{S \subseteq G : |S| = p^n\}$

Steps of Proof

Number Theoretic Section

1. From def of X to $|X| = \binom{p^n m}{p^n}$
2. $|X| \equiv m \pmod{p}$ by Lemma 3.3(ii)
3. $p \nmid |X|$

Orbit Section

4. Left multiplication is a group action
5. For each S , there exists $T \in \text{Orb}_G(S)$ such that $1_G \in T$
6. $\text{Orb}_G(S) = \text{Orb}_G(T)$ by Corollary 2.17(i)
7. List the distinct orbits in X $\{\text{Orb}_G(S_i) : i \in [1, t]\}$, assume $1_G \in S_i$
8. Orbits partition by Corollary 2.17(ii)
9. If $V_1, V_2, \dots, V_n$ are disjoint finite sets, then the disjoint union $V = V_1 \sqcup V_2 \sqcup \dots \sqcup V_n$ has $|V| = \sum_{i=1}^n |V_i|$
10. Rearrange/Assume $\{S_i\}$ is such that $p \nmid |\text{Orb}_G(S_i)|$ for $1 \leq i \leq r$ and $p \mid |\text{Orb}_G(S_i)|$ for $r < i \leq t$ for some arbitrary r .

Claim 1

11. If $g \in \text{Stab}_G(S_i)$ then $gS_i = S_i$ by definition
12. By assumption $1_G \in S_i$ then we have $g = g1_G \in gS_i$ Thus $\text{Stab}_G(S_i) \subseteq S_i$
13. By assumption $p \nmid |\text{Orb}_G(S_i)|$, also by Orbit-Stabiliser Theorem $|G| = |\text{Orb}_G(S_i)||\text{Stab}_G(S_i)|$
14. therefore $p^n \mid |\text{Stab}_G(S_i)|$
15. $|S_i| = p^n \leq |\text{Stab}_G(S_i)|$
16. $S_i \subseteq \text{Stab}_G(S_i)$ - - isn't necessary to state by the argument in 17
17. $S_i = \text{Stab}_G(S_i)$ - - by set theory (if $A \subseteq B, |A| \geq |B|$ then $A = B$)

Claim 2

18. Define a map $f : \{\text{Orb}_G(S_i) : 1 \leq i \leq r\} \rightarrow \text{Syl}_p(G)$ by $f(\text{Orb}_G(S_i)) = S_i$, this is well defined by Claim 1.
19. Let $1 \leq i, j \leq r$, and suppose that $S_i = S_j$. Then $\text{Orb}_G(S_i) = \text{Orb}_G(S_j)$
20. f is one to one
21. Let $P \in \text{Syl}_p(G)$, Then $P \in X$
22. $\text{Stab}_G(P) = P$ by Lemma 1.15
23. $p \nmid m = |G : P| = |G : \text{Stab}_G(P)| = |\text{Orb}_G(P)|$
24. $\text{Orb}_G(P) = \text{Orb}_G(S_i)$ for some $1 \leq i \leq r$
25. $P \in \text{Orb}_G(S_i)$
26. $P = gS_i$ for some $g \in G$
27. $1_G \in gS_i$
28. $g^{-1} \in S_i$
29. By Claim 1, $g \in S_i$
30. $gS_i = S_i$
31. $P = S_i = f(\text{Orb}_G(S_i))$ so f is onto
32. f is a bijection
33. $r = |\text{Syl}_p(G)|$

Conclusion

34. By Lemma, 3.3(ii), $|X| \equiv m \pmod{p}$
35. $|X| \equiv \sum_{i=1}^r |Orb_G(S_i)| \pmod{p}$
36. By Claim 1, $|Orb_G(S_i)| = |G : Stab_G(S_i)| = \frac{|G|}{|S_i|} = m$
37. By Claim 2, $r = |Syl_p(G)|$
38. $m \equiv mSyl_p(G) \pmod{p}$
39. As $p \nmid m$, m has an inverse u modulo p . Multiply by u .

Formalisation

Useful mathlib areas and lemmas proving hard to find

Useful pages

- Mathlib.Algebra.Group.End has stuff surrounding definitions of group actions
- MulAction.toPermHom says "Given an action of a group G on a set α , each $g : G$ defines a permutation of α ."
- Mathlib.GroupTheory.GroupAction.Blocks gives orbits partition results
- Mathlib.Algebra.Pointwise.Stabilizer has theorems about the stabiliser of a set of points
- MulAction.set_mem_fixedBy_iff says for a set S and $g \in G$, S is in the fixedBy set for g if $gS = S$ – could be useful for 12

$$s \in \text{fixedBy } (Set \alpha) \ g \iff \forall (x : \alpha), \ g \cdot x \in s \iff x \in s$$

Still needs to be decided/found:

- Need to look for in a Mathlib library: 1,3,4,5,7,12^a,24^b,25^c,26^d
- Need to see how we can do this in Lean: 7,10,18

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^asee MulAction.mem_stabilizer.set

^bunlikely to be an explicit theorem; it kind of comes from wrangling orbits partition, not really sure what to use

^csee MulAction.mem_orbit_self

^dCould come straight from definition of orbit? Depends on how the orbit is defined

Hard to find

- 1,5^e,
- ~~4 - we have to define ourselves I think; the structure of a group action is in there under SMul, and MulAut.conj defines the conjugacy action, so could try and edit MulAut.conj to define left multiplication - SMul works fine~~
- 7 - List, List.Mem apply to the list aspect, then the assumption $1_G \in S_i$) we write in as an assumption

Definitions

Definitions taken from Mathlib

- Group
- Subgroup
- Group Action? and therefore orbit & stabiliser
- Cardinality

Definitions we define

- Sylow Group

Lemmas

Lemmas taken from Mathlib

- 1 - Finset.powersetCard and Finset.card_powersetCard
- 3 - ZMod.natCast_eq_zero_iff
- 11? - rw as $g \in \text{Stab}_G(S_i) \implies \forall x \in S_i, g \cdot x \in S_i \leftrightarrow x \in S_i$
MulAction.set_mem_fixedBy_iff - says S is fixed by g iff $g \cdot x \in S \iff x \in S$
MulAction.mem_stabilizer_set - defines the stabiliser of a set – should suffice
MulAction.stabilizer_mul_self - I'm not entirely sure how it works but it seems related
- 12? - MulAction.mem_stabilizer_set - defines the stabiliser of a set – should suffice again?
- 14 - Prime.pow_dvd_of_dvd_mul_left
- 15 - Nat.le_of_dvd

^eWe can use an argument for by inverses, look at the actual proof for reference

- 16 - We don't need to state by our argument for 17
- 17 - `Finset.eq_of_subset_of_card_le` - have $Stab_G(S_i) \subseteq S_i$ by 12 and $|S_i| \leq |Stab_g(S_i)|$ by 15. Given these, `Finset.eq_of_subset_of_card_le` proves $S_i = Stab_G(S_i)$
- 19 - `orbit_subgroup_eq_self_of_mem`
- 23 - comes from orbit stabiliser ($p \nmid m$ by assumption $\gcd(p, m) = 1$, rest of equalities come from orbit stabiliser)
- 25 - `MulAction.mem_orbit_self` - not formatted for the orbit of sets but is the right idea to get $P \in Ord_G(P) = Orb_G(S_i)$
- 27 - `Subgroup.one_mem` gives $1_G \in P$
- 28 - `mem_leftCoset_iff`
- 30 - `leftCoset_mem_leftCoset`

Lemmas we prove

- Orbit-Stabiliser Theorem covers 13
- Lagranges Theorem
- Corollary 2.17 covers 6,8
- Number Theoretic - Lemma 3.3 covers 2 and 34
- Summation bits - 9 & 38
- 20,31,32,33 - Bijection Lemmas
- Lemma 1.15 covers 22
- Follows from definitions or what we have already shown in the proof: 21,29,35,36,37
- 39
- 12
- 3

Remaining

Merge current proofs

24-33

35-38

Provide reasonable comments and theorem names

Small project overview readme